Chemistry in Everyday Life

Question1

Given below are two statements:
Statement I : Noradrenaline is a neurotransmitter.
Statement II : Low level of noradrenaline is not the cause of depression in human.
In the light of the above statements, choose the correct answer from the options given below
[24-Jan-2023 Shift 1]

Options:

A. Statement I is correct but Statement II is incorrect

B. Statement I is incorrect but Statement II is correct

C. Both Statement I and Statement II are correct

D. Both Statement I and Statement II are incorrect

Answer: A

Solution:

Solution:
Fact

Question2

Match List I with List II

	LIST I Type		LIST II Name
A.	Antifertility drug	I.	Norethindrone
B.	Tranquilizer	II.	Meprobomate
C.	Antihistamine	III.	Seldane
D.	Antibiotic	IV.	Ampicillin

Choose the correct answer from the options given below:
[24-Jan-2023 Shift 2]

Options:

A. A-II, B-I, C-III, D-IV

B. A-IV, B-III, C-II, D-I

C. A-I, B-III, C-II, D-IV

D. A-I, B-II, C-III, D-IV

Answer: D

Solution:

Solution:
Theoretical, NCERT based.

Question3

Which of the following statements is incorrect for antibiotics? [25-Jan-2023 Shift 1]

Options:

A. An antibiotic must be a product of metabolism.

B. An antibiotic is a synthetic substance produced as a structural analogue of naturally occurring antibiotic.

C. An antibiotic should promote the growth or survival of microorganisms.

D. An antibiotic should be effective in low concentrations.

Answer: C

Solution:

Solution:
An antibiotic should not promote growth or survival of microorganisms. Antibiotics should inhibit growth of microbes.

Question4

Match List I with List II

List-I Antimicrobials	List-II Names
A Narrow SpectrumAntibiotic	I Furacin
B Antiseptic	II Sulphur dioxide
C Disinfectants	III Penicillin-G
D Broad spectrum antibiotic	IV Chloramphenicol

[29-Jan-2023 Shift 1]

Options:

A. (A) - III, (B) - I, (C) - II, (D) - IV

B. (A) - I, (B) - II, (C) - IV, (D) - III

C. (A) − II, (B) − I, (C) − IV, (D) - III

D. (A) − III, (B) − I, (C) − IV, (D) − II

Answer: A

Solution:

Solution:
(A) Narrow spectrum antibiotic - penicillin-G
(B) Antiseptic - Furacine
(C) Disinfectants - sulphur dioxide
(D) Broad spectrum antisiotics - chloramphenicol

Question5

A doctor prescribed the drug Equanil to a patient. The patient was likely to have symptoms of which disease?
[29-Jan-2023 Shift 2]

Options:

A. Stomach ulcers

B. Hyperacidity

C. Anxiety and stress

D. Depression and hypertension

Answer: D

Solution:

Solution:
Theory based.

Question6

Amongst the following compounds, which one is an antacid ?
[30-Jan-2023 Shift 1]

Options:

A. Ranitidine

B. Meprobamate

C. Terfenadine

D. Brompheniramine

Answer: A

Solution:

Solution:
1. Ranitidine: Antacid
2. Meprobamate: Tranquilizer
3. Terfenadine: Antihistamine
4. Brompheniramine: Antihistamine

Question7

Given below are two statements: One is labelled as Assertion A and the other labelled as Reason R.
Assertion A: Antihistamines do not affect the secretion of acid in stomach.
Reason R : Antiallergic and antacid drugs work on different receptors.
In the light of the above statements, choose the correct answer from the options given below:
[30-Jan-2023 Shift 2]

Options:

A. A is false but R is true

B. Both A and R are true and R is the correct explanation of A

C. A is true but R is false

D. Both A and R are true but R is not the correct explanation of A.

Answer: B

Solution:

Solution:
Antiallergic and antacid drugs work on different receptors
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Question8

Which of the following artificial sweeteners has the highest sweetness value in comparison to cane sugar?
[31-Jan-2023 Shift 1]

Options:

A. Aspartame

B. Sucralose

C. Alitame

D. Saccharin

Answer: C

Solution:

Solution:
Sweetness value order wrt cane sugar
Alitame > Sucralose > Saccharin > Aspartame

Question9

Which of the following compounds are not used as disinfectants?
(A) Chloroxylenol
(B) Bithional
(C) Veronal
(D) Prontosil
(E) Terpineol
Choose the correct answer from the options given below:
[31-Jan-2023 Shift 2]

Options:

A. A, B, E

B. A, B

C. B, D, E

D. C, D

Answer: A

Solution:

Solution:
Veronal is neurological medicine, Prontosil is antibiotic, rest all are disinfectants.

Question10

Match List I with List II

List I	List II
A. Tranquilizers	I. Anti blood clotting
B. Aspirin	II. Salvarsan
C. Antibiotic	III. Antidepressant drugs
D. Antiseptic	IV. Soframicine

Choose the correct answer from the options given below:
[1-Feb-2023 Shift 1]

Options:

A. (A) - IV, (B) - II, (C) - I, (D) - III

B. (A) - II, (B) - I, (C) - III, (D) - IV

C. (A) - III, (B) - I, (C) - II, (D) - IV

D. (A) - II, (B) - IV, (C) - I, (D) - III

Answer: C

Solution:

Solution:
NCERT (Chemistry in every day life)

Question11

Among the following, the number of tranquilizer/s is/are _____.
A. Chloroliazepoxide
B. Veronal
C. Valium
D. Salvarsan
[1-Feb-2023 Shift 2]

Answer: 3

Solution:

Solution:
(chlorodiazepoxide, Veronal, Valium is tranquilizer where as salvarsan is antibiotic.

Question12

Given below are two statements :-
Statement-I : Morphine is a narcotic analgesis. It helps in relieving pain without producing sleep.
Statement-II : Morphine and its derivatives are obtained from opium poppy.
In the light of the above statements, choose the correct answer from the options given below.
[6-Apr-2023 shift 2]

Options:

A. Statement I is true but Statement II is false

B. Both Statement I and Statement II are false

C. Both Statement I and Statement II are true

D. Statement I is false but Statement II is true

Answer: D

Solution:

Solution:
Statement-I- Morphine relieves in pain and produce sleep (incorrect)
Statement-II - Correct

Question13

Match List I with List II:←

List I	List II
A. Saccharin	I. High potency sweetener
B. Aspartame	II. First artificial sweetening agent
C. Alitame	III. Stable at cooking temperature
D. Sucralose	IV. Unstable at cooking temperature

Choose the correct answer from the options given below: [8-Apr-2023 shift 1]

Options:

A. A-II, B-III, C-IV, D-I

B. A-II, B-IV, C-I, D-III

C. A-IV, B-III, C-I, D-II

D. A-II, B-IV, C-III, D-I

Answer: B

Solution:

Solution:
(A) Saccharin → First artificial sweetening agent
(B) Aspartame → Unstable at cooking temperature used in soft drink and cold drink.
(C) Alitame → High potency sweetener (2000 more sweeter than cane sugar)
(D) Sucralose → Stable at coocking temperature. Also it does not provide calories.

Question14

The reaction used for preparation of soap from fat is : [10-Apr-2023 shift 2]

Options:

A. an addition reaction

B. an oxidation reaction

C. alkaline hydrolysis reaction

D. reduction reaction

Answer: C

Solution:

Solution:
The process of making is soap is saponification.
Ester + Base → Alcohol + Soap
In saponification, triglycerides are combine with strong base and form fatty acid so this is alkaline Hydrolysis reaction.

Question15

Which of the following is not a broad spectrum antibiotic? [24-Jun-2022-Shift-1]

Options:

A. Vancomycin

B. Ampicillin

C. Ofloxacin

D. Penicillin G

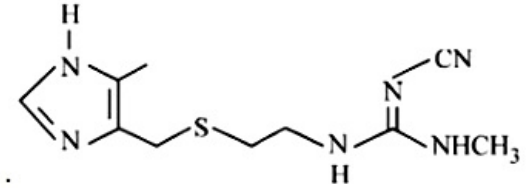
Answer: D

Solution:

Solution:
Penicillin G following is a narrow spectrum antibiotic

Question16

The structure shown below is of which well-known drug molecule?



[24-Jun-2022-Shift-2]

Options:

A. Ranitidine

B. Seldane

C. Cimetidine

D. Codeine

Answer: C

Solution:

Solution:
The given structure is that of cimetidine which is well known antacid.

Question17

Using very little soap while washing clothes, does not serve the purpose of cleaning of clothes, because:
[25-Jun-2022-Shift-1]

Options:

A. soap particles remain floating in water as ions.

B. the hydrophobic part of soap is not able to take away grease.

C. the micelles are not formed due to concentration of soap, below its CMC value.

D. colloidal structure of soap in water is completely disturbed.

Answer: C

Solution:

Solution:
The little amount of using soap while washing clothes, does not serve the purpose of cleaning of clothes because the micelles are not formed due to the low concentration of soap as it is below CMC.

Question18

Which one of the following is an example of artificial sweetner?
[25-Jun-2022-Shift-1]

Options:

A. Bithional

B. Alitame

C. Salvarsan

D. Lactose

Answer: B

Solution:

Solution:
Alitame is a second generation dipeptide sweetner that is 2000 times sweeter than sucrose.

Question19

Match List I with List II

List - I		List -II	
A.	Zymase	I.	Stomach
B.	Diastase	II.	Yeast
C.	Urease	III.	Malt
D.	Pepsin	IV.	Soyabean

Choose the correct answer from the options given below: [25-Jun-2022-Shift-2]

Options:

A. A-II, B-III, C-I, D-IV

B. A-II, B-III, C-IV, D-I

C. A-III, B-II, C-IV, D-I

D. A-III, B-II, C-I, D-IV

Answer: B

Solution:

Zymase naturally occurs in yeast.
Diastase is found in malt.
Urease is found in soyabean
Pepsin is found in stomach

Question20

An antiseptic dettol is a mixture of two compounds ' A ' and ' B ' where A has 6π electrons and B has 2π electrons. What is 'B'? [25-Jun-2022-Shift-2]

Options:

A. Bithionol

B. Terpineol

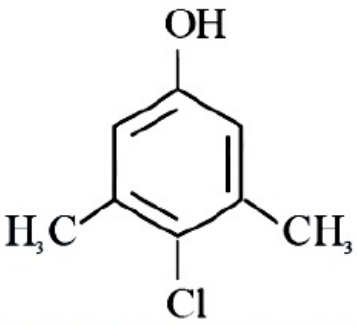
C. Chloroxylenol

D. Chloramphenicol

Answer: B

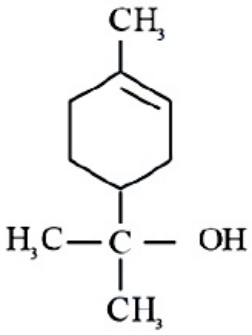
Solution:

Dettol is mixture of



Chloroxylenol (Compound A) It has $6\pi e^-$

and



Terpineol (Compound B) It has $2\pi e^-$

Question21

A chemical which stimulates the secretion of pepsin is: [26-Jun-2022-Shift-1]

Options:

A. Anti histamine

B. Cimetidine

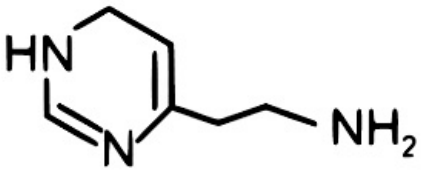
C. Histamine

D. Zantac

Answer: C

Solution:

Solution:
Histamine stimulates the secretion of pepsin and hydrochloric acid in the stomach.



Histamine

Question22

The measured BOD values for four different water samples (A − D) are as follows: A = 3 ppm; B = 18 ppm; C = 21 ppm; D = 4 ppm. The water samples which can be called as highly polluted with organic wastes, are [26-Jun-2022-Shift-2]

Options:

A. A and B

B. A and D

C. B and C

D. B and D

Answer: C

Solution:

Solution:
Highly polluted water should have BOD value of 17 ppm or more

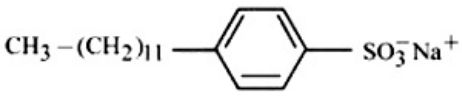
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Question23

Which of the following is NOT an example of synthetic detergent? [26-Jun-2022-Shift-2]

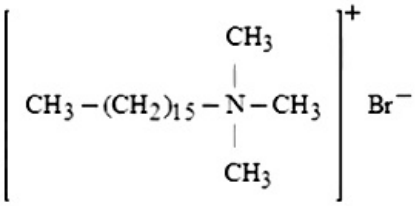
Options:

A.



B. $CH_3 - (CH_2)_{16} - COO^- Na^+$

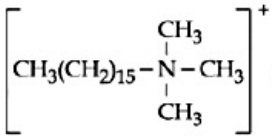
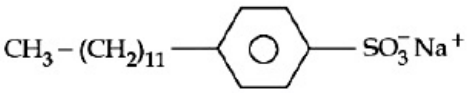
C.



D. $CH_3(CH_2)_{16}COO(CH_2CH_2O)_nCH_2CH_2OH$

Answer: B

Question24

	List - I		List - II
(A)		(I)	Dishwashing powder
(B)		(II)	Toothpaste
(C)	$C_{17}H_{35}COO^-Na^+ + Na_2CO_3$ + Rosinate	(III)	Laundry soap
(D)	$CH_3(CH_2)_{16}COO(CH_2CH_2O)_nCH_2CH_2OH$	(IV)	Hair conditioner

Choose the correct answer from the options given below : [27-Jun-2022-Shift-1]

Options:

A. (a) (A) − (III), (B) − (II), (C) − (IV), (D) − (I)

B. (b) (A) − (IV), (B) − (II), (C) − (III), (D) − (I)

C. (c) (A) − (IV), (B) − (III), (C) − (II), (D) − (I)

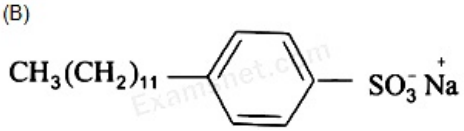
D. (d) (A) − (III), (B) − (IV), (C) − (I), (D) − (II)

Answer: B

Solution:

(A) $[CH_3(CH_2)_{15} - N(CH_3)_3]^+Br^-$ is cationic detergents used in hair conditioner

(B)



is anionic detergent used in tooth pastes.

(C) $C_{17}H_{35}COO^{-}Na^{+} + Na_2CO_3$ + Rosin ate is used as laundary soap
(D) $CH_3(CH_2)_{16}COO(CH_2CH_2O)_N CH_2 CH_2 OH$ is nonionic detergents formed from stearic acid and poly ethylene glycol used as liquid dishwashing detergents

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Question25

Choose the correct answer from the options given below :

	List-I		List-II
(A)	Antipyretic	I.	Reduces pain
(B)	Analgesic	II.	Reduces stress
(C)	Tranquilizer	III.	Reduces fever
(D)	Antacid	IV.	Reduces acidity (stomach)

[27-Jun-2022-Shift-2]

Options:

A. A-III, B-I, C-II, D-IV

B. A-III, B-I, C-IV, D-II

C. A-I, B-IV, C-II, D-III

D. A-I, B-III, C-II, D-IV

Answer: A

Solution:

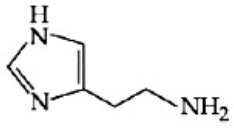
Solution:
Antipyretic - Reduces fever
Analgesic - Reduces pain
Tranquilizer - Reduces stress
Antacid - Reduces Acidity (stomach)

Question26

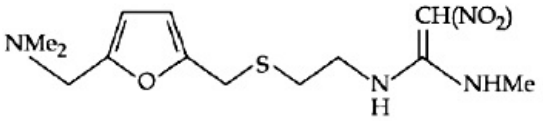
The drug tegamet is :
[28-Jun-2022-Shift-2]

Options:

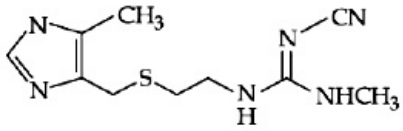
A.



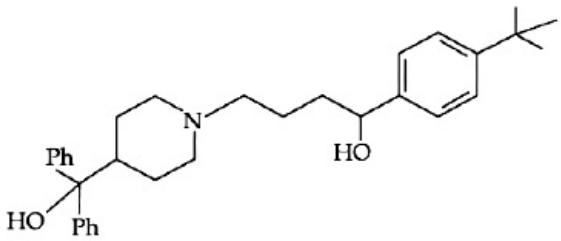
B.



C.



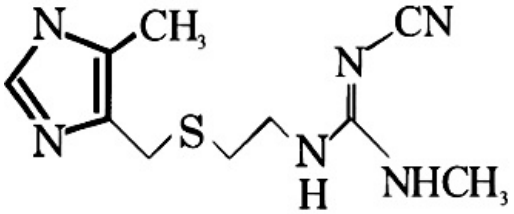
D.



Answer: C

Solution:

Solution:
Tegamet is the brand name of Cimetidine



Question27

Which of the following compound does not contain sulphur atom? [29-Jun-2022-Shift-1]

Options:

A. Cimetidine

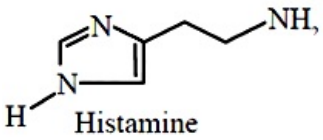
B. Ranitidine

C. Histamine

D. Saccharin

Answer: C

Solution:



Histamine

Histamine is nitrogenous compound it does notcontain sulpher

Question28

Drugs used to bind to receptors, inhibiting is natural function and blocking a message are called:
[25-Jul-2022-Shift-1]

Options:

A. Agonists

B. Antagonists

C. Allosterists

D. Anti histaminists

Answer: B

Solution:

Solution:
Drugs that bind to the receptor site and inhibit its natural function are called Antagonists.

Question29

List I	List II
(A) Sulphate	(I Pesticide
(B) Fluoride	(II) Bending of bones
(C) Nicotine	(III) Laxative effect
(D) Sodium arsinite	(IV) Herbicide

Choose the correct answer from the options given below:
[25-Jul-2022-Shift-2]

Options:

A. A-II, B-III, C-IV, D-I

B. A-IV, B-III, C-II, D-I

C. A-III, B-II, C-I, D-IV

D. A-III, B-II, C-IV, D-I

Answer: C

Solution:

Sodium arsinite - Herbicide
Nicotine - Pesticide
Sulphate - Laxative effect
Fluoride - Bending of bones

Question30

Some drugs bind to a site other than the active site of an enzyme. This site is known as
[25-Jul-2022-Shift-2]

Options:

A. non-active site

B. allosteric site

C. competitive site

D. therapeutic site

Answer: B

Solution:

Solution:
Some drugs do not bind to the enzyme’s active site. These bind to a different site of enzyme which is called allosteric site.

Question31

Stearic acid and polyethylene glycol react to form which one of the following soap/s detergents ?
[26-Jul-2022-Shift-1]

Options:

A. Cationic detergent

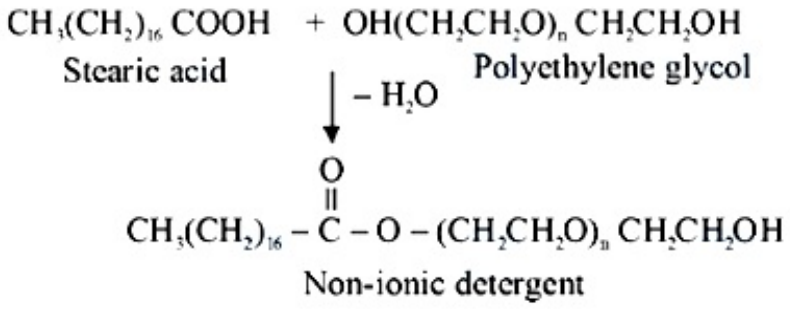
B. Soap

C. Anionic detergent

D. Non-ionic detergent

Answer: D

Solution:



Question32

How many of the following drugs is/are example(s) of broad spectrum antibiotics?
Ofloxacin, Penicillin G, Terpineol, Salvarsan.
[26-Jul-2022-Shift-2]

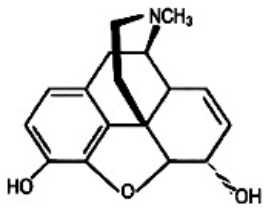
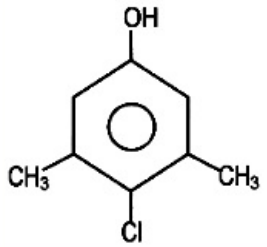
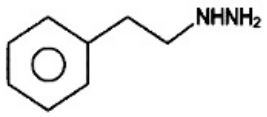
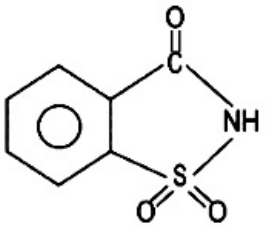
Answer: 1

Solution:

Ofloxacin is the only broad-spectrum antibiotic given in the question
Penicilin - G is a narrow-spectrum antibiotic.
Salvarsan is mainly active against spirochete, a bacteria that causes syphilis
Terpineol is an antiseptic.

Question33

Match List I with List II.

List I	List II
(A) 	(I)Anti-depressant
(B) 	(II)550 times sweeter than cane sugar.
(C) 	(III)Narcotic analgesic
(D) 	(IV)Antiseptic

Choose the correct answer from the options given below: [27-Jul-2022-Shift-1]

Options:

A. A-IV, B-III, C-II, D-I

B. A-III, B-I, C-II, D-IV

C. A-III, B-IV, C-I, D-II

D. A-III, B-I, C-IV, D-II

Answer: C

Solution:

Solution:
A is morphine which is a narcotic analgesic.
B is chloroxylenol, an antiseptic.
C is Nardil, an antidepressant.
D is saccharin, which is around 550 times sweeter than cane sugar.

Question34

Which of the following enhances the lathering property of soap? [27-Jul-2022-Shift-2]

Options:

A. Sodium stearate

B. Sodium carbonate

C. Sodium rosinate

D. Trisodium phosphate

Answer: C

Solution:

Solution:
A gum called rosin is added to soap which forms sodium rosinate. It helps to produce lather.

Question35

Two statements in respect of drug-enzyme interaction are given below
Statement I : Action of an enzyme can be blocked only when an inhibitor blocks the active site of the enzyme.
Statement II : An inhibitor can form a strong covalent bond with the enzyme.
In the light of the above statements, choose the correct answer from the options given below
[28-Jul-2022-Shift-2]

Options:

A. Both Statement I and Statement II are true

B. Both Statement I and Statement II are false

C. Statement I is true but Statement II is false

D. Statement I is false but Statement II is true

Answer: D

Solution:

Action of an enzyme can be altered by a number of factors like temperature, pH, presence of activators and coenzymes and presence of inhibitors and poisons.
Inhibitors or poisons interact with the active functional groups on the enzyme surface and often reduce or completely destroy the catalytic activity of the enzymes.

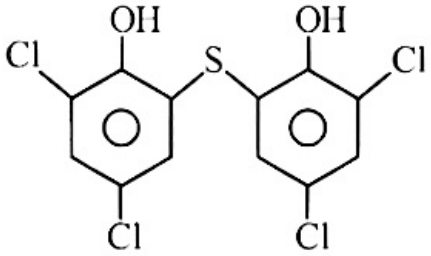
Question36

The number of chlorine atoms in bithionol is _____.
[29-Jul-2022-Shift-2]

Answer: 4

Solution:

Solution:
Number of chlorine atoms in bithionol = 4



Question37

Match List-I with List-II

List-I	List-II
A. Be	1. Treatment of cancer
B. Mg	2. Extraction of metals
C. Ca	3. Incendiary bombs and signals
D. Ra	4. Windows of X-ray tubes
	5. Bearings for motor engines.

Choose the most appropriate answer the option given below.
[18 Mar 2021 Shift 2]

Options:

A. A- 4 ,B- 3,C- 1,D- 2

B. A- 4,B- 3,C- 2,D- 1

C. A- 3,B- 4,C- 5,D- 2

D. A- 3,B- 4,C- 2,D- 5

Answer: B

Solution:

Solution:
(a)Be-(iv)windows of X-ray tubes

Motor engine bearings are used, Bearings can be made smaller and lighter because of the superior strength copper-beryllium has with the alternative bronze landing gear material.
(b) Mg -(iii)incendiary bombs and signals
It combines with water to produce gas and oxygen. Both the powdered and solid forms of magnesium are employed as incendiary agents for light and antipyretic reasons.
(c) Ca-(ii)extraction of metals
Pure calcium metal is employed as a reducing agent when producing other metals such as thorium, uranium and zirconium.
(d)Ra-(i)treatment of cancer
Ra- is used in cancer treatment because it is radioactive.

Question38

Match List-I with List-II.

List-I (Class of chemicals)	List-II (Example)
A. Antifertility drug	1. Meprobamate
B. Antibiotic	2. Alitame
C. Tranquilizer	3. Norethindrone
D. Artificial sweetener	4. Salvarsan

[18 Mar 2021 Shift 2]

Options:

A. A-2 ,B- 3,C- 4,D- 1

B. A- 4,B- 3,C- 2,D- 1

C. A- 3,B- 4,C- 1,D- 2

D. A- 2,B- 4,C- 1,D- 3

Answer: C

Solution:

(A) Antifertility drug → Norethindrone
Norethindrone is a synthetic progesterone which is amongst the most commonly used antifertility drugs.
(B) Antibiotic → Salvarsan
Arsphenamine, also known as salvarsan used as antibiotic.
(C) Tranquilizer → Meprobamate
Meprobamate is used to treat anxiety disorders or for short-term relief of the symptoms of anxiety in adults and children 6 years of age and older. Meprobamate is in a class of medications called tranquilizers.
(D) Artificial sweetener → Alitame
Alitame is an artificial sweetener containing aspartic acid.
Hence, correct match is (A-3), (B-4), (C-1), (D-2).

Question39

Match the list-I with list- II

List-I	List-II
(A) Chlorophyll	(i) Ruthenium
(B) Vitamin $-B_{12}$	(ii) Platinum
(C) Anticancer drug	(iii) Cobalt
(D) Grubbs catalyst	(iv) Magnesium

Choose the most appropriate answer from the options given below [18 Mar 2021 Shift 1]

Options:

A. A - 3, B - 2, C - 4, D - 1

B. A - 4, B - 3, C - 2, D - 1

C. A - 4, B - 3, C - 1, D - 2

D. A - 4, B - 2, C - 3, D - 1

Answer: B

Solution:

Solution:
A. Chlorophyll is coordination complex of Mg.
B. Vitamin B_{12}, cyanocobalamine is a coordination compound of cobalt.
C. cis-platin $[Pt(NH_3)_2Cl_2]$ used in the treatment of cancer and is a coordination compound of platinum.
D. Grubbs catalyst is a series of catalysts containing ruthenium.
∴ Correct match is (A-iv), (B-iii), (C-ii), (D-i)

Question40

Match the List-I with List-II.

List-I (Class of drug)	List-II (Example)
(A) Antacid	(i) Novestrol
(B) Artificial sweetener	(ii) Cimetidine
(C) Antifertility	(iii) Valium
(D) Tranquilizers	(iv) Alitame

[18 Mar 2021 Shift 1]

Options:

A. A - 2, B - 4, C - 1, D -3

B. A - 4, B - 1, C - 2, D -3

C. A - 4, B - 3, C - 1, D -2

D. A - 2, B - 4, C - 3, D -1

Answer: A

Solution:

A. Antacid - Cimetidine Cimetidine is a stomach acid reducer that is used to treat and prevent certain types of stomach ulcer.
B. Artificial sweetener-Alitame Alitame is an aspartic acid containing dipeptide sweetener.
C. Antifertility - Novestrol Antifertility drugs is a chemical substances which suppress the action of hormones that promote pregnancy. An antifertility drug is novestrol (ethynylestradiol). Another example of antifertility drugs is norethindrone.
D. Tranquilizers - Valium Valium, trade name of a tranquiliser drug introduced by the pharmaceutical company Hofmann-La-Roche in 1963.
∴ Correct match is A-(ii), B-(iv), C-(i), D-(iii)

Question41

With respect to drug-enzyme interaction, identify the wrong statement [17 Mar 2021 Shift 1]

Options:

A. Non-competitive inhibitor binds to the allosteric site.

B. Allosteric inhibitor changes the enzyme's active site.

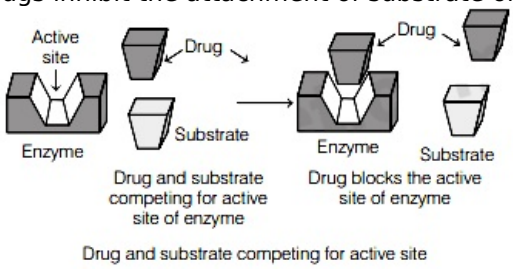
C. Allosteric inhibitor competes with the enzyme's active site.

D. Competitive inhibitor binds to the enzyme's active site.

Answer: C

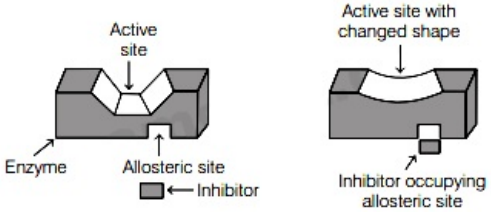
Solution:

Solution:
Allosteric inhibitor combines at allosteric site and not the active site.
Drugs inhibit the attachment of substrate on active site of enzymes in two different ways;



Drug and substrate competing for active site

(ii) Some drugs do not bind to the enzyme's active site. These bind to a different site of enzyme which is called allosteric site. This binding of inhibitor at allosteric site changes the shape of the active site in such a way that substrate cannot recognize it. This is also called non-competitive inhibitor.



Question42

Match List-I and List-II.

List-I (Chemical compound)	List-II(Used as)
A. Sucralose	1. Synthetic detergent
B. Glyceryl ester of	2. Artificial stearic acid sweetener
C. Sodium benzoate	3. Antiseptic
D. Bithionol	4. Food preservative

Choose the correct match.
[17 Mar 2021 Shift 2]

Options:

A. A- 4 ,B- 3,C- 2,D- 1

B. A- 2,B- 1,C- 4,D- 3

C. A- 3,B- 2,C- 4,D- 1

D. A- 1,B- 2,C- 4,D- 3

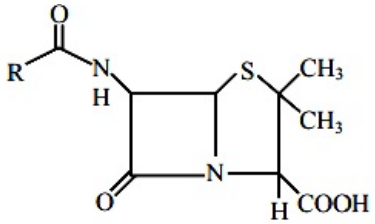
Answer: B

Solution:

Solution:
The correct matches are A − 2, B − 1, C − 4, D − 3.
A. Sucralose → Artificial sweetener Artificial sweetener is the sugar that is sweet like sugar but has zero or low calories. Artificial sweeteners can be plant extract or chemical synthesised. Saccharin, aspartame, sodium cyclamate and sucralose are the examples of artificial sweeteners.
B. Glyceryl ester of stearic acid → Synthetic detergent Esters of stearic acid are used in synthesis of soaps and detergents.
C. Sodium benzoate → Food preservative The food is oxidised or hydrolysed by several factors such as air that is known as aerial oxidation, or by moisture, bacteria and light. Sodium benzoate has an aromatic ring that can stabilise the free radical which causes the oxidation of food. Sodium benzoate is a food preservative.
D. Bithionol → Antiseptic Bithionol is added to soap to impart antiseptic property. It reduces odour produced by bacterial decomposition of organic matter on the skin.

Question43

Given below are two statements :
Statement I : Penicillin is a bacteriostatic type antibiotic.
Statement II : The general structure of Penicillin is:



Choose the correct option :
[27 Jul 2021 Shift 2]

Options:

A. Both statement I and statement II are false

B. Statement I is incorrect but statement II is true

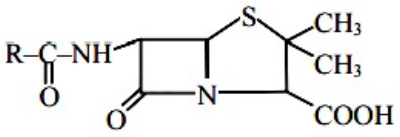
C. Both statement I and statement II are true

D. Statement I is correct but statement II is false

Answer: B

Solution:

Solution:
Statement I : Pencillin is bactericidal not bacteriostatic hence given statement is false.
Statement II : Structure of pencilline given is correct



Question44

Match List-I with List-II :

	List-I (Drug)		List-II (Class of Drug)
(a)	Furacin	(i)	Antibiotic
(b)	Arsphenamine	(ii)	Tranquilizers
(c)	Dimetone	(iii)	Antiseptic
(d)	Valium	(iv)	Synthetic antihistamines

Choose the most appropriate match : [27 Jul 2021 Shift 1]

Options:

A. (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

B. (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

C. (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

D. (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

Answer: D

Solution:

Solution:
→ furacine acts as Antiseptic
→ Arsphenamine also known as salvarsan acts as antibiotic

→ Dimetone is synthetic histamine
→ valium is a Tranqulizer

Question45

Which one of the following chemical agent is not being used for dry-cleaning of clothes?
[25 Jul 2021 Shift 1]

Options:

A. H_2O_2

B. CCl_4

C. Liquid CO_2

D. $Cl_2C = CCl_2$

Answer: A

Solution:

Solution:
CO_2, CCl_4 and $Cl_2C = CCl_2$ are used as dry cleaning agents for clothes.
H_2O_2 is used as bleaching agent in laundry.
Carbon tetrachloride was commonly blended with other solvents for use as a dry cleaning solvent.
Now, because of its high toxicity and tendency to corrode equipment, carbon tetrachloride is no longer used as a drycleaning solvent.

Question46

Sodium stearate $CH_3(CH_2)_{16}COO^-Na^+$ is ananionic surfactant which forms micelles in oil.
Choose the correct statement for it from thefollowing:
[25 Jul 2021 Shift 1]

Options:

A. It forms spherical micelles with $CH_3(CH_2)_{16}-$ group pointing towards the centre of sphere.

B. It forms non-spherical micelles with $-COO^-$ group pointing outwards on the surface.

C. It forms spherical micelles with $CH_3(CH_2)_{16}-$ group pointing outwards on the surface of sphere

D. It forms non–spherical micelles with $CH_3(CH_2)_{16}-$group pointing towards the centre.

Answer: A

Solution:

Forms spherical micelles with $CH_3(CH_2)_{16}$ group pointing towards the centre of sphere

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Question47

Green chemistry in day–to–day life is in the use of:
[20 Jul 2021 Shift 1]

Options:

A. Chlorine for bleaching of paper

B. Large amount of water alone for washing clothes

C. Tetrachloroethene for laundry

D. Liquified CO_2 for dry cleaning of clothes

Answer: D

Solution:

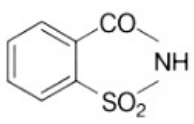
Solution:
Chlorine gas was used earlier for bleaching paper. These days, hydrogen peroxide (H_2O_2) with suitable catalyst.
Tetra chlroroethene ($Cl_2C = CCl_2$) was earlier used as solvent for dry cleaning. The compound contaminates the ground water and is also a suspected carcinogen. Replacement of halogenated solvent by liquid CO_2 will result in less harm to groundwater.
Hence given statement (4) is correct.

Question48

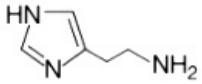
Which one of the following chemicals is responsible for the production of HCl in the stomach leading to irritation and pain?
[27 Aug 2021 Shift 2]

Options:

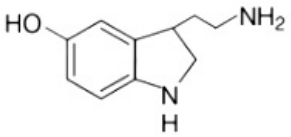
A.



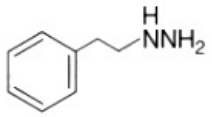
B.



C.



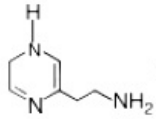
D.



Answer: B

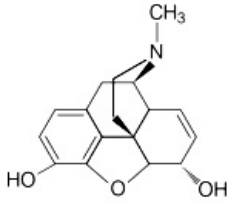
Solution:

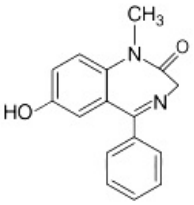
Solution:

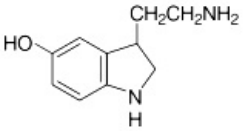
The compound  is responsible for the production of HCl in the stomach leading to irritation and pain.
So, the option (b) is correct.

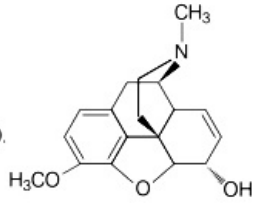
Question49

Consider the following structures :

A. 

B. 

C. 

D. 

The correct statement about (A), (B), (C) and (D) is [27 Aug 2021 Shift 1]

Options:

A. (A), (B) and (C) are narcotic analgesics

B. (B), (C) and (D) are tranquillizers

C. (A) and (D) are tranquillizers

D. (B) and (C) are tranquillizers

Answer: D

Solution:

Solution:
The structures A and D are narcotic analgesics, where A is morphine and D is codeine.
The structures B and C are tranquillizers viz. diazepam and serotonin respectively.

Question50

A chemist has 4 samples of artificial sweetener A, B, C and D. To identify these samples, he performed certain experiments and noted the following observations:
(I) A and D both form blue-violet colour with ninhydrin.
(II) Lassaigne extract of C gives positive $AgNO_3$ test and negative $Fe_4[Fe(CN)_6]_3$ test.
(III) Lassaigne extract of B and D gives positive sodium nitroprusside test.
Based on these observations which option is correct?
[Jan. 09,2020(I)]

Options:

A. A: Aspartame; B: Saccharin; C: Sucralose; & D: Alitame

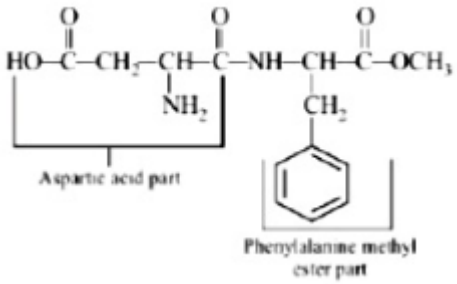
B. A: Alitame; B: Saccharin; C: Aspartame; D: Sucralose

C. A: Saccharin; B: Alitame; C: Sucralose; D : Aspartame

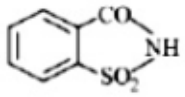
D. A: Aspartame; B: Alitame; C: Saccharin; D: Sucralose

Answer: A

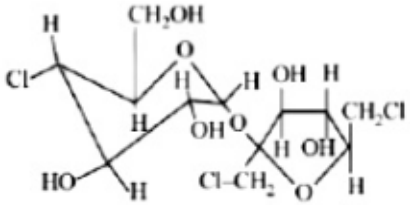
Solution:



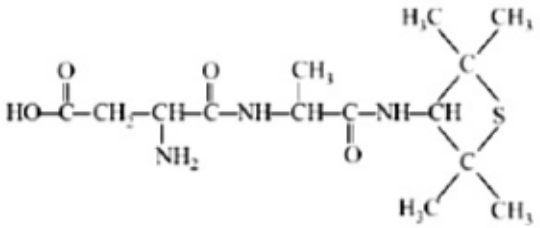
B – Saccharine



C – Sucralose



D – Alitame



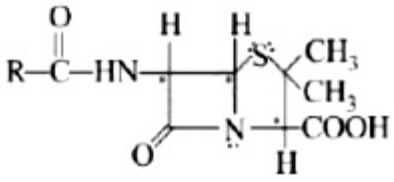
(I) A and D give positive test with ninhydrin because both have free carboxylic and amine groups.
(II) C form precipitate with $AgNO$, in the lassaigne extract of the sugar because it has chlorine atoms.
(III) B and D give positive test with sodium nitroprusside because both have sulphur atoms.

Question51

The number of chiral centres in penicillin is _____.
[NV Jan.08,2020(I)]

Answer: 3

Solution:



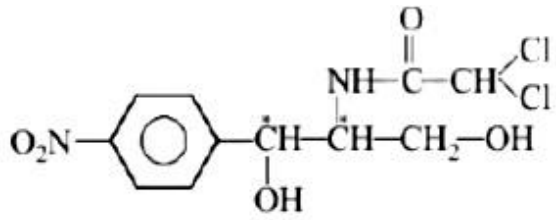
There are three chiral centres in penicillin.

Question52

The number of chiral carbons in chloramphenicol is _____.
[NV Jan. 07, 2020 (I)]

Answer: 2

Solution:



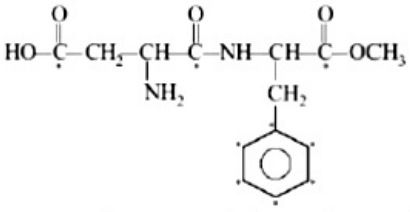
Question53

The number of sp^2 hybridised carbons present in "Aspartame" is _____. [NV Jan. 07,2020 (II)]

Answer: 9

Solution:

Solution:



Structure of aspartame is shown above. It is a methyl ester of dipeptide formed from aspartic acid and phenylalanine. sp^2 hybridised carbon atoms are shown by the star mark in the structure.

Question54

If a person is suffering from the deficiency of nor-adrenaline, what kind of drug can be suggested? [Sep. 05, 2020 (I)]

Options:

A. Anti-inflammatory

B. Antidepressant

C. Antihistamine

D. Analgesic

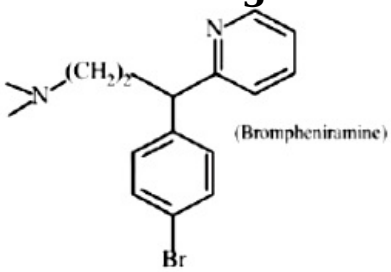
Answer: B

Solution:

Solution:
If the level of noradrenaline is low, then the person suffers from depression. In such situations, anti-depressant drugs are required.

Question55

The following molecule acts as an:



[Sep. 0 5, 2020(II)]

Options:

A. Antiseptic

B. Anti-depressant

C. Anti-bacterial

D. Anti-histamine

Answer: D

Solution:

Solution:
Synthetic drugs, brompheniramine (dimetapp) acts as antihistamines.

Question56

The mechanism of action of "Terfenadine" (Seldane) is:
[Sep. 04, 2020(II)]

Options:

A. Activates the histamine receptor

B. Inhibits the secretion of histamine

C. Helps in the secretion of histamine

D. Inhibits the action of histamine receptor

Answer: D

Solution:

Seldane is an anti-histamine drug that has inhibitory action on histamine receptor.

Question57

The antifertility drug "Novestrol" can react with : [Sep .03,2020(I)]

Options:

A. $ZnCl_2$ / HCl; $FeCl_3$; Alcoholic HCN

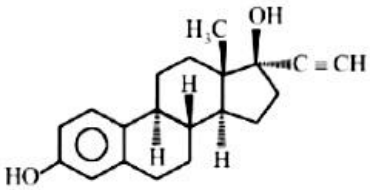
B. Br_2/ water: $ZnCl_2$ / HCl; $FeCl_3$

C. Alcoholic HCN; NaOCl; $ZnCl_2$ / HCl

D. Br_2/ water; $ZnCl_2$ / HCl; NaOCl

Answer: B

Solution:



Novestrol has phenolic, alcoholic and terminal alkyne groups, so it can react with Br_2 water, $ZnCl_2$ / HCl as well as $FeCl_3$

Question58

Match the following drugs with their therapeutic actions:

(i) Ranitidine	(A) Antidepressant
(ii) Nardil(Phenelzine)	(B) Antibiotic
(iii) Chloramphenicol	(C) Antihistamine
(iv) Dimetane	(D) Antacid (Brompheniramine)
	(E) Analgesic

[Sep. 03,2020 (II)]

Options:

A. (i)-(A);(ii)-(C); (iii)-(B); (iv)-(E)

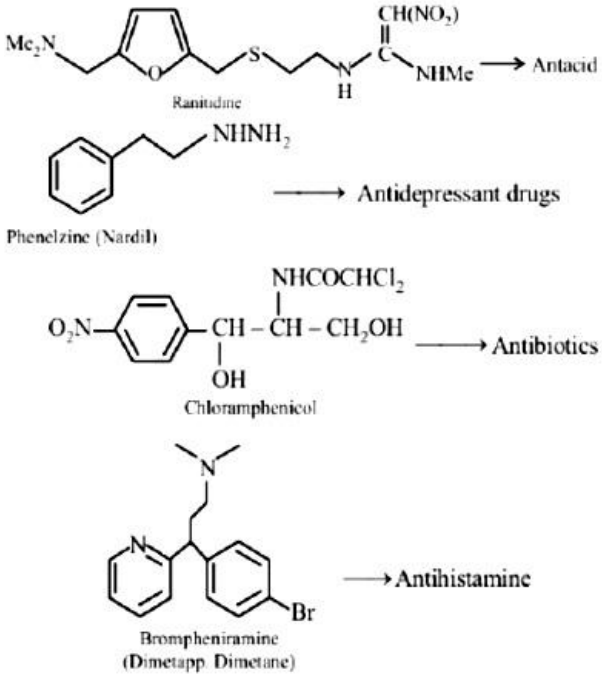
B. (i)-(D); (ii)-(A);(iii)-(B); (iv)-(C)

C. (i)-(E); (ii)-(A); (iii)-(C);(iv)-(D)

D. (i)-(D);(ii)-(C); (iii)-(A); (iv)-(E)

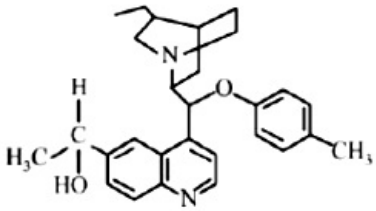
Answer: B

Solution:



Question59

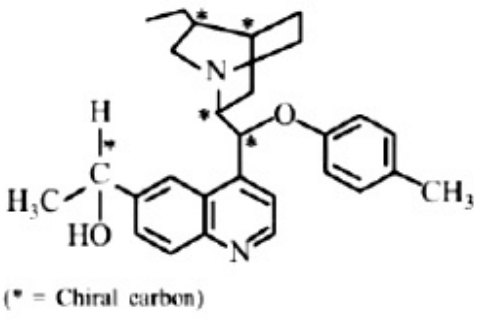
The number of chiral carbons present in the molecule given below is _____.



[NV Sep. 02, 2020 (I)]

Answer: 5

Solution:



(* = Chiral carbon)

Question60

Noradrenaline is a/an:
[April 9, 2019 (II)]

Options:

A. Antacid

B. Neurotransmitter

C. Antidepressant

D. Antihistamine

Answer: B

Solution:

Solution:
Noradrenaline, also called Norepinephrine, is a neurotransmitter. It plays a major role in mood change.

Question61

The correct match between item (I) and item (II) is:

Item - I	Item - II
(A) Norethindrone	(P) Antibiotic
(B) Ofloxacin	(Q) Antifertility
(C) Equanil	(R) Hypertension
	(S) Analgesics

[Jan.11,2019(I)]

Options:

A. (A) → (Q); (B) → (R); (C) → (S)

B. (A) → (Q); (B) → (P); (C) → (R)

C. (A) → (R); (B) → (P); (C) → (S)

D. (A) → (R); (B) → (P); (C) → (R)

Answer: B

Solution:

Solution:
(a) **Norethindrone** is a progesterone derivative, so it is an antifertility drug (Q).
(b) **Ofloxacin** is an antibiotic (P).
(c) **Equanil** is a mild tranquilizer and hence used in hypertension (R).

Question62

The correct match between Item I and Item II is:

Item - I	Item - II
(A) Allosteric effect	(P) Molecule binding to the active site of enzyme
(B) Competitive inhibitor	(Q) Competitive communication in the body
(C) Receptor	(R) Molecule binding to a site other than the active site of enzyme
(D) Poison	(S) Molecule binding to the enzyme covalently

[Jan. 11, 2019 (II)]

Options:

A. (A) → (R); (B) → (P); (C) → (Q); (D) → (S)

B. (A) → (P); (B) → (R); (C) → (Q); (D) → (S)

C. (A) → (R); (B) → (P); (C) → (S); (D) → (Q)

D. (A) → (P); (B) → (R); (C) → (S); (D) → (Q)

Answer: A

Solution:

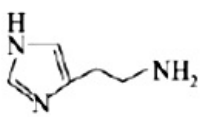
Solution:
Allosteric → Molecule bind to a site other than the active site of enzyme
Competitive inhibitor→ Molecule bind to the active site inhibitor of enzyme
Receptor → Molecule crucial for communication in the body.
Poison → Molecule binding to the enzyme covalently

Question63

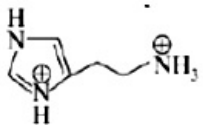
The predominant form of histamine present in human blood is (pK_2, Histidine -6.0)
[2018]

Options:

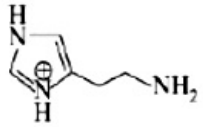
A.



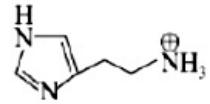
B.



C.



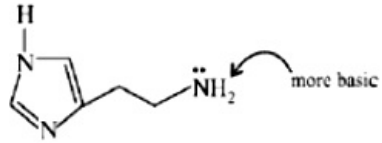
D.



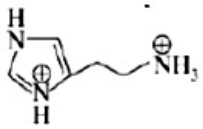
Answer: D

Solution:

Solution:
Structure of histamine



Blood is slightly basic in nature ($7.35 \mathrm{pH}$). At this pH, terminal $\mathrm{NH_2}$ will get protonated due to more basic nature.
∴ Predominant structure of histamine is



Question64

The correct match between items of List-I and List-II is:

List-I	List-II
(A) Phenelzine	(p) Pyrimidine
(B) Chloroxylenol	(q) Furan
(C) Uracil	(r) Hydrazine
(D) Ranitidine	(s) Phenol

[Online April 16, 2018]

Options:

A. (A) − (s); (B) − (r); (C) − (q); (D) − (p)

B. (A) − (r); (B) − (s); (C) − (p); (D) − (q)

C. (A) − (r); (B) − (s); (C) − (q); (D) − (p)

D. (A) − (s); (B) − (r); (C) − (p); (D) − (q)

Answer: B

Solution:

Solution:
→ Phenelzine contains hydrazine
→ Chloroxylenol contains phenol
→ Uracil is the pyrimidine base
→ Ranitidine contains furan ring

Question65

The reason for "drug induced poisoning" is : [Online April 8,2017]

Options:

A. Binding reversibly at the active site of the enzyme

B. Bringing conformational change in the binding site of enzyme

C. Binding irreversibly to the active site of the enzyme

D. Binding at the allosteric sites of the enzyme

Answer: C

Solution:

Solution:
Enzyme inhibition can be either reversible or irreversible. In case of irreversible inhibition the inhibitor dissociates very slowly from its target enzyme as it has tightly bound to the enzyme either covalently or non-covalently. Some irreversible inhibitors are important drugs like Penicilline and Aspirin. Thus "drug induced poisoning" may bound irreversibly to the active site of the enzyme.

Question66

Which of the following is an anionic detergent? [2016]

Options:

A. Cetyltrimethyl ammonium bromide.

B. Glyceryl oleate.

C. Sodium stearate.

D. Sodium lauryl sulphate.

Answer: D

Solution:

Sodium lauryl sulphate ($C_{11}H_{23}CH_2OSO_3Na^+$) is an anionic detergent. Glyceryl oleate is a glyceryl ester of oleic acid. Sodium stearate ($C_{17}H_{35}COO^-Na^+$) is a soap. Cetyltrimethyl ammonium bromide $\left[CH_3(CH_2)_{15}\overset{+}{N}(CH_3)_3\right]Br^-$ is a cationic detergent.

Question67

Which of the following is a bactericidal antibiotic? [Online April 10,2016]

Options:

A. Ofloxacin

B. Tetracycline

C. Chloramphenicol

D. Erythromycin

Answer: A

Solution:

Solution:
Bactericidal are the drugs that kills bacteria. Ofloxacin works by stopping the growth of bacteria. This antibiotic treats only bacterial infections.

Question68

The artificial sweetener that has the highest sweetness value in comparison to cane sugar is: [Online April 9,2016]

Options:

A. Sucralose

B. Aspartane

C. Saccharin

D. Alitame

Answer: D

Solution:

Solution:
Alitame is an artificial sweetner that is 2,000 times as sweet as sugar.

Question69

Which of the following compounds is not an antacid? [2015]

Options:

A. Phenelzine

B. Ranitidine

C. Aluminium hydroxide

D. Cimetidine

Answer: A

Solution:

Solution:
Phenelzine is an antidepressant, while others areantacids.

Question70

Which artificial sweetener contains chlorine? [Online April 11, 2015]

Options:

A. Sucralose

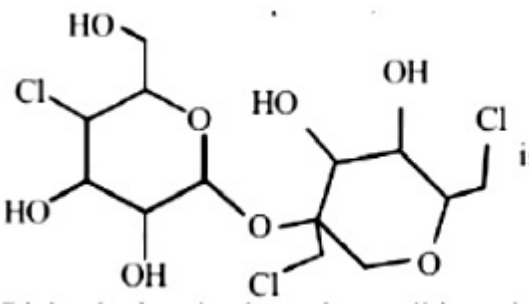
B. Alitame

C. Aspartame

D. Saccharin

Answer: A

Solution:



is sucralose.

Question71

Which one of the following is used as Antihistamine?

[Online April 11, 2014]

Options:

A. Omeprazole

B. Chloranphenicol

C. Diphenhydramine

D. Norethindrone

Answer: C

Solution:

Solution:
Diphenhydramine is used as antihistamine.

Question72

Aminoglycosides are usually used as:
[Online April 12,2014]

Options:

A. antibiotic

B. analgesic

C. hypnotic

D. antifertility

Answer: A

Solution:

Solution:
The aminoglycosides are among the oldest antibiotics.

Question73

Which of the following statements about aspirin is not true?
[Online April 22, 2013]

Options:

A. It is effective in relieving pain.

B. It is a neurologically active drug.

C. It has antiblood clotting action.

D. It belongs to narcotic analgesics.

Answer: D

Solution:

Aspirin is an non-narcotics analgesic.

Question74

H_1− Receptor antagonists is a term associated with : [Online April 23, 2013]

Options:

A. Antiseptics

B. Antihistamins

C. Antacids

D. Analgesics

Answer: B

Solution:

Solution:
The term "antihistamine" refers only to H_1 antagonists, which is also known as H_1,-receptor antagonists and H_1 -antihistamine.

Question75

Aspirin is known as: [2012]

Options:

A. Acetyl salicylic acid

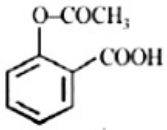
B. Phenyl salicylate

C. Acetyl salicylate

D. Methyl salicylic acid

Answer: A

Solution:



Aspirin (Acetyl salicylic acid)

Question76

Which one of the following compounds is an anti-fertility drug? [Online May 7, 2012]

Options:

A. Aspirin

B. Chloromycetin

C. Saheli

D. Penicillin

Answer: C

Solution:

Solution:
The antifertility drugs are those chemical compounds which prevent pregnancy in woman. These, drugs controls the menstrual cycle and ovulation and so used as birth control drug or contraceptives.

Question77

Chemically heroin is [Online May 12,2012]

Options:

A. morphine dibenzoate

B. morphine monoactate

C. morphine diacetate

D. morphine monobenzoate

Answer: C

Question78

Aspirin can be prepared by the reaction of [Online May 19,2012]

Options:

A. Salicyldehyde with aceticanhydride in presence of H_2SO_4

B. Salicylic acid with methanol in presence of H_2SO_4

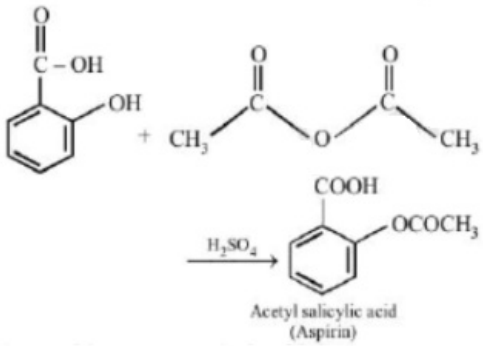
C. Salicylic acid with acetic anhydride in presence of H_2SO_4

D. Cinnamic acid with acetic anhydride in presence of H_2SO_4

Answer: C

Solution:

Solution:
Asprin is prepared by reaction of salicylic acid with acetic anhydride in presence of H_2SO_4.



Question79

Sulphonamides act as [Online May 26,2012]

Options:

A. Antiseptic

B. Analgesic

C. Antimicrobials

D. Antipyretic

Answer: C

Solution:

Solution:
Sulphonamides act as antimicrobials.

Question80

Which one of the following types of drugs reduces fever? [2005]

Options:

A. Tranquiliser

B. Antibiotic

C. Antipyretic

D. Analgesic

Answer: C

Solution:

Solution:
An antipyretic is a drug which is responsible for lowering the temperature of the feverish body to normal but has no effect on normal temperature states.

Question81

Which of the following could act as a propellant for rockets? [2003]

Options:

A. Liquid oxygen + liquid argon

B. Liquid hydrogen + liquid oxygen

C. Liquid nitrogen + liquid oxygen

D. Liquid hydrogen + liquid nitrogen

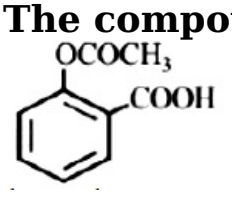
Answer: B

Solution:

Solution:
Liquid hydrogen and liquid oxygen are used as excellent fuel for rockets. $H_2(\ell)$ has low mass and high enthalpy of combustion, whereas oxygen is a strong supporter of combustion.

Question82

The compound



is used as
[2002]

Options:

A. antiseptic

B. antibiotic

C. analgesic

D. pesticide.

Answer: C

Solution:

The given compound is aspirin which is antipyretic and analgesic.
